DATA DICTIONARY

Formula 1 World Championship Dataset

1950 – 2022 | 4 Tables | Power BI Hackathon

1. Dataset Overview

The dataset contains **25,840 race result records** spanning 73 seasons of Formula 1 (1950–2022). It is split into 4 separate CSV files designed to be loaded into Power BI and connected as a **Snowflake Schema**.

Table	Role	Rows	Key Columns
fact_Results	Fact Table	25,840	result_id · season_year · round_number · driver_name · team · position · points
dim_Races	Dimension	1,079	season_year · round_number · grand_prix · race_date · circuit_name
dim_Drivers	Dimension	855	driver_name · nationality · date_of_birth · race_number
dim_Circuits	Dimension	76	circuit_name · location · country · lat · lng · altitude

2. Relationships (Snowflake Schema)

All three relationships use **text-based join keys** — not numeric IDs. This means **TRIM and case consistency** in Power Query is critical before creating relationships.

From (Many)	To (One)	Join Keys	Cardinality
fact_Results	dim_Races	season_year + round_number	Many → One (Composite Key)
fact_Results	dim_Drivers	driver_name	Many → One
dim_Races	dim_Circuits	circuit_name	Many → One

⚠ **dim_Circuits connects to dim_Races — not to fact_Results. To use circuit data in a measure, you must navigate through dim_Races first using RELATED().**

3. Table Definitions

fact_Results → 25,840 rows rows

Core fact table. One row per driver per race. Contains all race outcomes, points, lap data, and qualifying times.

F1 Dataset — Data Dictionary | 4 Tables | 25,840 Total Records | 1950 – 2022

Column Name	Data Type	Nulls	Unique	Description	Sample Values
result_id PK	PK	✓ None	25,840	Unique identifier for each result record.	RES00001
season_year FK	Integer	✓ None	73	The F1 season year. Used as part of composite FK to dim_Races.	1950 … 2022
round_number FK	Integer	✓ None	22	Race round within the season. Combined with season_year to uniquely identify a race.	1, 2, 3 … 22
driver_name FK	Text	✓ None	855	Full driver name. FK to dim_Drivers. Must match exactly after TRIM. ⚠ **Extra whitespace may cause relationship failure — always TRIM this column.**	Max Verstappen
team	Text	✓ None	210	Constructor/team name for this result. Spans 73 years of team history.	Red Bull, Ferrari
grid	Integer	✓ None	35	Starting grid position (qualifying order).	1, 2, 3 … 34
position	Text	✓ None	39	Finishing position. Numeric = classified finish. R = Retired/DNF. F = Did Not Start. ⚠ **MUST clean before use. Create conditional column: if R or F → null, else cast to Integer.**	1, R, F, 14
points	Decimal	✓ None	39	Championship points awarded. Stored as decimal — change type to Integer. ⚠ **Stored as text decimal (e.g. 9.0). Change Type → Integer in Power Query.**	25.0, 18.0, 0.0
laps	Integer	✓ None	172	Number of laps completed by the driver.	70, 58, 44
total_time	Text	18,752	6,839	Race time. Winner gets absolute time, others get gap to winner. 72.6% null (DNF or not recorded). ⚠ **High null rate is expected and historical — do not drop this column.**	2:13:23.6, +52.0
fastest_lap	Time	18,461	6,772	Driver's fastest lap time. Only recorded from 2004 onwards. ⚠ **71.4% null — historical. Pre-2004 data had no fastest lap tracking.**	1:24.125
fl_rank	Integer	18,249	26	Rank among all drivers for fastest lap in that race. 1 = fastest.	1, 2, 3 … 20
notes	Text	✓ None	137	Reason for result. 'Finished' = completed race. Others are DNF reasons (Engine, Gearbox, Collision…).	Finished, Engine, Accident
qualifying_pos	Integer	16,265	29	Overall qualifying position. Stored as float — change to Integer. ⚠ **Empty before ~2003. Expected — qualifying format changed over the decades.**	1.0, 4.0, 14.0

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Column Name	Data Type	Nulls	Unique	Description	Sample Values
q1_time	Time	16,412	8,434	Q1 session lap time. Empty before 2003.	1:16.290
q2_time	Time	20,676	4,879	Q2 session lap time. Empty when driver did not advance past Q1.	1:28.289
q3_time	Time	22,659	3,072	Q3 session (pole shootout) lap time. Only top 10 drivers participate.	1:31.702

dim_Races → 1,079 rows rows

One row per race event. Contains race date, circuit, and season info. Bridges fact_Results to dim_Circuits.

Column Name	Data Type	Nulls	Unique	Description	Sample Values
season_year PK	Integer	✓ None	73	Season year. Part of composite PK with round_number.	1950 … 2022
round_number PK	Integer	✓ None	22	Race round within season. Part of composite PK with season_year.	1, 2 … 22
grand_prix	Text	✓ None	53	Official race name. 53 unique Grand Prix names across 73 seasons.	British Grand Prix
race_date	Date	✓ None	1,079	Full date of the race. Stored as text — must be converted to Date type. ⚠ Change Type → Date in Power Query. Required for all Time Intelligence measures.	1950-05-13
circuit_name FK	Text	✓ None	76	Circuit name. FK to dim_Circuits. Must match exactly.	Silverstone Circuit
country	Text	✓ None	34	Host country of the race.	UK, Italy, Monaco

dim_Drivers → 855 rows rows

One row per driver. Covers all 855 drivers who competed in F1 from 1950 to 2022.

Column Name	Data Type	Nulls	Unique	Description	Sample Values
driver_name PK	Text	✓ None	855	Full driver name. PK of this table. Must match fact_Results[driver_name] exactly. ⚠ Always TRIM both sides of the relationship. Case must also match.	Lewis Hamilton
nationality	Text	✓ None	42	Driver's country of nationality. 42 unique nationalities.	British, Dutch, Italian
date_of_birth	Date	✓ None	837	Driver date of birth in YYYY-MM-DD format. Use to calculate Age at race time.	1985-01-07

Column Name	Data Type	Nulls	Unique	Description	Sample Values
race_number	Integer	2	99	Official car number assigned to the driver. 2 nulls exist (historical drivers had no fixed number).	44, 1, 33, 16

dim_Circuits → 76 rows rows

One row per circuit. Contains geographic coordinates enabling map visuals. Connected to dim_Races — not directly to fact_Results.

Column Name	Data Type	Nulls	Unique	Description	Sample Values
circuit_name PK	Text	✓ None	76	Official circuit name. PK of this table. FK from dim_Races[circuit_name].	Silverstone Circuit
location	Text	✓ None	74	City or locality where the circuit is located.	Silverstone, Monaco
country	Text	✓ None	34	Country where the circuit is located.	UK, Italy, Monaco
lat	Float	✓ None	76	Latitude coordinate. Use with lng for Map visual in Power BI.	52.0786, 43.7347
lng	Float	✓ None	76	Longitude coordinate. Use with lat for Map visual in Power BI.	-1.01694, 7.42056
altitude	Integer	2	66	Circuit altitude in meters above sea level. 2 nulls. ⚠ Replace null with 0 or average altitude.	153, 7, 551

4. Known Data Quality Issues — ETL Checklist

All issues below are **intentional** and part of the grading. Fix them in Power Query before building relationships or DAX measures.

Column	Issue	Affected Rows	Recommended Fix
Multiple columns	\N stored as null string	~18,000+	Replace Values: '\N' → null
position	Mixed: '1','R','F' in same col	8,805 = R 1,368 = F	Conditional Column: R/F → null, else Int
points	Decimal stored as Text (9.0)	~25,840	Change Type → Integer (removes .0)
race_date	Text, not Date type	1,079	Change Type → Date
driver_name	Extra whitespace in some names	~handful	Transform → Trim
qualifying_pos	Float stored as Text (2.0)	~9,575	Change Type → Integer
altitude (Circuits)	2 missing values	2	Replace null → 0 or mean

Column	Issue	Affected Rows	Recommended Fix
`race_number (Drivers)`	2 missing values	2	`Replace null → 0`
`Q1/Q2/Q3 times`	Empty before 2003	~16,000+	`Expected — historical data, keep as null`
`total_time`	72.6% null	18,752	`Expected — only winner has absolute time`

5. Important Notes for Power BI

Composite Key: The relationship between fact_Results and dim_Races uses TWO columns: season_year and round_number together. In Power BI, create this as a composite relationship by Ctrl+clicking both columns in Manage Relationships.

RELATED() path: To access dim_Circuits from a measure on fact_Results, navigate: fact_Results → dim_Races → dim_Circuits. Use RELATED(dim_Races[circuit_name]) first, then chain.

Date table: For Time Intelligence (YTD, SAMEPERIODLASTYEAR), mark dim_Races[race_date] as a Date column and build a Calendar table using CALENDAR() or CALENDARAUTO().

Position filter: Always filter out 'R' and 'F' values from position before counting finishers. A driver who retired is NOT in position 0.

Historical nulls: The large null counts in Q1/Q2/Q3 and fastest_lap are NOT errors — they reflect real historical gaps in F1 data collection before 2003-2004.